

Feeding flocks on rangelands: insights into the local ecological knowledge of shepherds in Boulemane province (Morocco)

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Abstract. In Mediterranean regions, traditional pastoral systems involve shepherds leading flocks along daily grazing circuits on arid rangelands. Over the past decades, these systems have become increasingly agro-pastoral and the importance given to feeding flocks on rangelands is variable. Our study aimed at investigating the local ecological knowledge (LEK) about forage plants and animal foraging behaviour of shepherds in a pastoral area of Morocco, and eventually analysing the possible interactions between such LEK, its utilisation for grazing management and the pastoral status of the farm. Eleven semi-directive interviews with shepherds, either salaried or owning their own farm, were carried out at three sites differing in terms of agricultural context and available forage resources. Shepherds' LEK included recognising and naming forage plants and rangeland types, identifying preferred or less preferred plants or plant parts, describing circumstantial palatability of plants depending mainly on season, other locally available plants and watering times. LEK about animal feeding preferences and its integration into grazing management was more extensive at sites where pastoral systems were still most valued, and for shepherds who were either experienced or who were considering the activity in the future. Conversely, young salaried shepherds or farmer-shepherds who devoted more attention to the agricultural component of their system seemed to be less knowledgeable about the subject. In a context where pastoralism is challenged both by the higher profitability of agriculture and by the depletion of pastoral resources as a result of frequent droughts and decreased surface area devoted to grazing lands, the future of such LEK is uncertain. The perpetuation of LEK might depend on the ability of local extension services to value farmers' LEK and to help them enrich it with scientific knowledge.

Additional keywords: grazing management, indigenous knowledge, pastoral industry, rangeland pasture.

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Introduction

In Mediterranean regions, rangelands are widely used as a source of grazed fodder, especially in farming systems that produce meat (Hadjigeorgiou *et al.* 2005; Jouven *et al.* 2010; Bernués *et al.* 2011). North Africa has extensive rangelands, ~168 million hectares (Boussaïd *et al.* 2004), which were traditionally used for small ruminant rearing in pastoral systems (Sidahmed 1996). Such pastoral systems are strategically important for a variety of reasons, ranging from the production of meat and milk to feed the local populations, to the conservation of landscapes and wildlife habitats (Blondel 2006; Hubert *et al.* 2014) and to the provision of income for local, often marginalised communities (FAO 2001). Over the past decades, North Africa's traditional pastoral systems have been replaced by agro-pastoral systems where animal feeding partially relies on purchased inputs (especially feed and drugs), and the activity of maintaining livestock has become less valued than cropping or non-agricultural activities (Bourbouze 2000). The increase in the use

of concentrate feed can be attributed to: (i) the intensification of livestock farming systems supported by agricultural policies in order to meet the growing demand in meat; and (ii) to the decrease in forage availability due to recurrent drought. Today, farming systems with a pastoral component can be highly variable in terms of the diversity of forage resources used, their strategic importance in the feeding system and their vulnerability to climatic hazards and socioeconomic changes (Tunisia: Jemaa *et al.* 2016). Changes can be expected in the traditional tasks carried out by shepherds and in the associated skills and knowledge, especially the local ecological knowledge (LEK) associated with rangeland utilisation.

LEK can be defined as '[a body of] biophysical observations, skills and technologies as well as social relationships (...) that structure human-environmental interactions' (Fernandez-Gimenez 2000). LEK is transferred from one generation to the next within a social group but also modified by individuals as a result of their own experiences and observations. LEK is

mobilised in order to produce suitable management practices and, in turn, is amended as a function of the observed consequences of the management practices applied. In a context where the social structures associated with pastoral farming are dissolving and shepherding is less valued, LEK about plants and grazing animals might be weakened or even disappear.

The objective of our study was to: (i) characterise the LEK of Moroccan shepherds in relation to the local agricultural context; and (ii) investigate how the management practices implemented could be connected to such knowledge.

Materials and methods

The area of study

The study was conducted in the north-eastern part of the Middle Atlas mountains (Fig. 1), which is a traditional area of small ruminant pastoral farming, although the importance and profitability of pastoral systems has decreased in recent years. The area has experienced the following changes over the past decades: (i) the expansion of crop land and olive orchards at the expense of pasture; (ii) the increase of the proportion of concentrate feed in the diet; and (iii) the scarcity of a specialised labour force for shepherding associated with a lack of family replacement for farmers (Chattou 2014; Boughalmi *et al.* 2015). Our work covered the province of Boulemane which, due to its agroecological diversity, can be considered as representative of the steppe zones of Morocco and western Algeria, but also of the eastern Middle Atlas Mountains.

We investigated three sites, which we will refer to by the name of the closest town: Boulemane, El Ksabi and Guigou (Fig. 1). Each site differs in terms of agricultural context and available forage resources. Boulemane, at an altitude of ~1800 m a.s.l., is one of the rare places in Morocco where it snows in winter. The topography is quite even and the dominant land use is steppe rangelands. Sheep farming is the main agricultural activity within the site and the associated farming systems are mainly pastoral. El Ksabi, at an altitude of ~1500 m a.s.l., is a relatively remote area with an irrigated type of farming. Orchards, cereal fields, steppe rangeland and livestock coexist in the area and sheep farming systems are agropastoral. Guigou, at an altitude

of ~1500 m a.s.l., is surrounded by low hills covered with cedar and pine forests and rain-fed type of farming with large areas cultivated with cereals and onions. The sheep farming systems here can be classified as agrosylvopastoral. The sites were considered to be different enough in their local agricultural context so as to provide us with a variety of cases in terms of type and amount of LEK associated with rangeland utilisation.

At all three sites, sheep farmers often raise a few goats. These goats may be either mixed with the sheep flock or kept separately. Although the sheep or mixed flocks are usually shepherded by men, small goat flocks are watched by women and children and kept close to the houses that are guarded by dogs. Goats are mainly kept for family consumption and they also complement sheep by browsing more frequently on shrubs and trees. Our study focussed on sheep, although the information about goats spontaneously provided by the shepherds is reported as well.

Choice of shepherds to be interviewed

We decided to carry out a limited number of comprehensive and in-depth interviews instead of a larger number of more superficial ones, in order to put the shepherd in the position of an expert sharing his individual LEK with the interviewer. To avoid suspicion or canned answers, we devoted adequate time before the start of the study *per se* to acquaint ourselves with local conventions and customs so as to achieve a better integration and acceptance within the community. Technicians from ANOC (*Association Nationale Ovine et Caprine*) acted as a contact point in the region, introducing us to local farmers. ANOC is a non-governmental organisation working with sheep and goat farmers in order to promote genetic selection of breeds and sanitary management and to organise the meat product chains (ANOC 2013) in order to satisfy the growing demand. ANOC technicians arranged short interviews with 35 sheep farmers whom we met individually or in groups at the farmers' homes and in the presence of a local representative who was a member of either the farmers' association or the village council. The prime rule was to avoid being associated with any local or national authority, and specific emphasis was laid on the solely scientific motivations of the work.

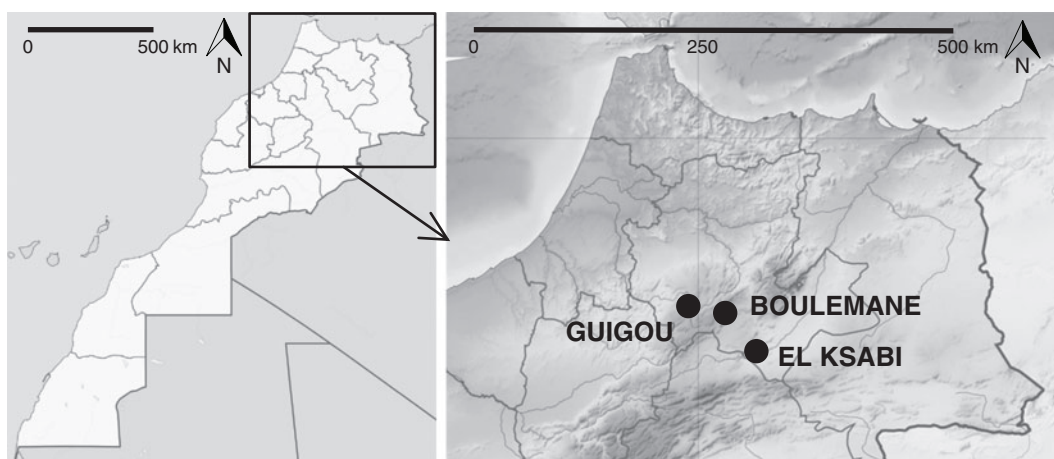


Fig. 1. Map of Morocco showing the three sites investigated in our study, each identified by the name of the closest town (credits for base map: Eric Gaba – Wikimedia Commons).

Three types of stakeholders were identified: farmers, farmer-shepherds and salaried shepherds. Farmers own large flocks consisting of several hundred sheep but due to their age or other activities, they cannot herd the animals. They thus employ salaried shepherds to implement the management strategy they have devised. Farmer-shepherds own the flock they care for and are thus free to make all the related decisions. They sometimes employ salaried shepherds to help them herd the animals. Salaried shepherds are only responsible for the herding of the flock and are generally not involved in other decisions related to the sheep farming activity. They usually follow the farmer's instructions about daily grazing circuits and related herding tactics. They sometimes own a small flock with several dozen sheep themselves.

Among the 35 stakeholders encountered, we selected 10 shepherds among the salaried shepherds and farmer-shepherds, corresponding to the systems with the highest pastoral component in each area and to the individuals with the highest propensity to communicate. An old farmer-shepherd incidentally encountered in the Boulemane region and very motivated by the study was added to the sample, increasing the number of shepherds interviewed to 11. This sampling procedure enabled us to identify 11 LEK holders within the main pastoral area of Morocco. Although the sample was not designed to be representative of the local population, considering three small regions that differed in their agricultural dynamics made it possible to capture the diversity of the situations reported in Boughalmi *et al.* (2015).

Shepherds' interviews

Eleven semi-structured interviews were carried out with the shepherds selected, with the aim of collecting their LEK about rangelands and animal foraging selectivity, and eventually connecting it to their herding practices. The method chosen to gather LEK was the semi-directive interview where the interviewer guides the discussion without interfering with the interviewee's train of thought (Yin 1994; Kauffman 1996). This method has been widely applied to collect LEK (Brook and McLachlan 2008) and other forms of experiential knowledge about how to graze animals on rangeland (Meuret and Provenza 2015a, 2015b). Semi-directive interviews have two major interests for our study: on the one hand, the conversational context is appropriate when participants are not comfortable with direct questions or might not understand them well and, on the other hand, it provides an opportunity to collect useful but unpredictable information (Huntington 2000).

Shepherds were interviewed *in situ* during the months of May–June 2013, corresponding to the end of spring and the beginning of summer, while they were herding their flocks. Interviews lasted from 3 h to a whole day as utmost care was taken to minimise disruptions to the herding activity. Interviews were carried out in the language of the shepherd, which was usually Berber. Asma Boughalmi, a doctoral student carrying out field measurements in the same region at the time, acted as an interpreter.

The interviews were intentionally not linear so as to have the most casual conversation possible rather than a formal interrogation. Making the shepherd feel that what he said was

of great value was key to our approach. We capitalised on the adaptation of semi-structured interviews for pastoral studies of Meuret (2014). The conversation with the shepherd was based on four main topics: (i) previous experience as a shepherd; (ii) flock characteristics in terms of composition and management; (iii) plants or associations of plants considered as forage by the shepherd and management practices implemented to use such plants (season of utilisation, design of daily grazing circuits, and so on); and (iv) observed foraging selectivity of the sheep in terms of plant consumption across feeding sites and seasons, eventually in relation to grazing management. Each topic was documented as follows: first, the motivation for that part of the interview; then, the main question formulated in simple terms to introduce the topic to the interviewee; and finally, a set of booster questions to provide the interviewer with alternative requests and stimulate the discussion. Despite our preliminary interviews to select the shepherds, the eagerness of interviewees to talk about rangeland plants and sheep behaviour was variable. Thus, long interviews (at least 3 h in a row) and a diversity of booster questions were necessary to gather sufficient information. The questions themselves were adapted to avoid antagonism or clichéd responses and thus ensure the reliability of the information collected.

A photo sequence of plants was included in all the interviews, usually towards the end, with the objective to confirm or supplement the information previously gathered. A series of 24 photographs of plant species expected to be found in the local rangelands were shown on an LCD screen or on a print scaled to the size of an A4 paper. The photographs showed whole plants or plant parts: young shoots, mature leaves, inflorescences. Plants considered as important by the shepherd and especially those spontaneously mentioned during the interview were shown first. The shepherd was asked whether he identified the plant, how he perceived its abundance and distribution in the rangeland and if he considered the plant or parts of it, at given stages or seasons, to be a relevant feed resource for his flock.

The conversations were audio recorded and used to produce an accurate transcription of what the shepherd said, providing data for analysis and comparisons. Recording such information both in writing and on a voice recorder gave a sense of importance to the words of the interviewees, a situation to which they are not accustomed. The fact that the objective of the interview was to gather information about plants but not sensitive information such as herd size, selling price or risk of overgrazing and damaging the woods, contributed to making the encounters cordial. These situations ultimately created an atmosphere that was conducive to collecting confidential information and reliable observations on the part of the interviewees despite the relatively heavy nature of the interviews due to the language barrier and the complexity of the subject.

Analysis of interviews and shepherds' LEK

A thematic analysis of the interviews was performed to characterise the shepherds, their practices and their LEK. On the basis of this analysis, we built three qualitative variables, with three levels each, to describe the type and amount of LEK for each shepherd and its integration into grazing management. These variables were used to determine 'profiles of LEK' among

the shepherds interviewed. The typology of LEK we applied was the following:

- (1) Knowledge about edible plants: level 1 = recognises the most abundant plants, but without naming them spontaneously; level 2 = recognises and spontaneously names the most abundant plants; level 3 = recognises and spontaneously names most plants if not all of them.
- (2) Knowledge about animal feeding preferences: level 1 = knowledge of preferred and less preferred plants or plant parts; level 2 = knowledge of preferred and less preferred plants in relation to the season; level 3 = knowledge of preferred and less preferred plants in relation to the season, watering and plant associations.
- (3) Integration of LEK into grazing management: level 1 = no such declaration; level 2 = within the daily grazing circuits, search for temporal associations of complementary plants in order to make the animals eat enough; level 3 = within daily and seasonal grazing circuits, sequencing of complementary rangeland sectors and their respective mixtures of plants.

Plant sampling and analysis

In order to attempt a connection between sheep foraging selectivity as reported by the shepherds and the chemical composition of the plants or plant parts ingested, one to four samples of the most cited plants or plant parts consumed by the animals were collected *in situ*. The plant samples were immediately placed in plastic bags and kept in a refrigerator (4°C), then transferred to the plant laboratory at the Institut Agronomique et Vétérinaire Hassan II and submitted to the botanists to validate the botanical genus. In Rabat samples were oven-dried until constant weight was reached (~95% dry weight, obtained in ~48 h), and then ground to a size of 1 mm. The 40 samples collected were sent to CIRAD (Montpellier, France) for analysis.

NIRS analysis was performed on a NIRSystem 6500 spectrometer (FOSS, Laurel, MD, USA) in reflectance mode from 400 nm to 2500 nm. Samples were scanned in duplicate and averaged. Chemical composition was predicted for each sample on the basis of its NIRS spectrum using existing calibrations (CIRAD, unpubl. data). The parameters considered in the calibrations were: total minerals (Ash: ashing at 550°C); crude protein (CP, Kjeldahl method); fibre fractions (neutral detergent fibre: NDF; acid detergent fibre: ADF; acid detergent lignin: ADL; Van Soest *et al.* 1991); and *in vitro* enzymatic organic matter digestibility (IVOMD; Aufrère *et al.* 2007). All values

were expressed as % dry matter, except for IVOMD that was expressed as % organic matter. Tannin content was assessed by two methods: total phenolic content was measured by colourimetry (Folin Ciocalteu reagent) and expressed as tannic acid equivalent (% DM), and condensed tannins were determined by vanillin assay and expressed as catechin equivalents (% DM).

Results

Due to the limited number of interviews, the results are not presented using percentages. Instead, we refer to individual cases with letters (A to K) or to the relative number of shepherds within the study area ($n/11$) or each sub-region ($n/5$ for Boulemane or $n/3$ for Guigou and El Ksabi). For results about LEK, we sometimes quote the interviewees in order to report their views as accurately as possible. We chose to keep, in quotes, the local names used to characterise rangeland and rangeland plants, together with their assumed Latin or English equivalent. These local names may have a slightly different meaning depending on the area as well as on the experience of the shepherd and his ability to distinguish plants with close physical similarities. For example, 'halfa' (*Stipa tenacissima*) and 'smar' (*Juncus maritimus*) were sometimes confused, although experienced shepherds referred to 'smar' as 'false halfa'.

The local farming context

Characteristics of the shepherds interviewed

All the shepherds interviewed had begun this activity in their childhood and were males (Table 1). Almost all shepherds in Morocco are males, and no shepherdesses were encountered in the area. Dogs were not involved in shepherding. They were instead kept close to the shed/house to discourage intruders and protect women and children.

The farmer-shepherds interviewed were assisted by a son or by a salaried shepherd who was occasionally entrusted with the flock while the farmer devoted time to meeting with other farmers, purchasing feed or selling animals. For example, B was accompanied during the interview by his 30-year-old son who had been entrusted with the flock and was thus often asked by his father to fill in the gaps. Salaried shepherds could be family or clan members and, in all of the cases, belonged to the local community. For example, C was alternatively assisted by a salaried shepherd during the summer and by his brother during the winter transhumance. Among the interviewees, G (full-time shepherd) and I (seasonal worker: shepherd in the

Table 1. Characteristics of the shepherds interviewed and flocks owned by salaried shepherds

Site Interviewee	Boulemane				Guigou				El Ksabi		
	A	B	C	D	E	F	G	H	I	J	K
Status (Sh: salaried shepherd; F-Sh: farmer-shepherd)	Sh	F-Sh	F-Sh	Sh	Sh	Sh	Sh	F-Sh	Sh	F-Sh	Sh
Age (approx.)	30	60	60	40	35	25	45	40	35	35	40
Relationship to the owner	brother	—	—	same clan	none	son	none	—	none	—	none
Sheep owned out of the flock (for Sh only)	0	0	0	150	50	0	50	0	"a few" ^A	0	50

^AThis salaried shepherd owned a few sheep, kept in a family flock (together with his brother's and father's sheep).

grazing season, working in the orchards in winter) were used to changing employers every year, their choice being driven by the remuneration offered. Salaried shepherds managed the flock alone. The few sheep or goats owned were kept as a separate entity and usually entrusted to family members, even to 8- to 10-year-old children who would learn the job and perhaps become farmer-shepherds one day. If close relatives such as the brother or father of the salaried shepherd also owned sheep, then a 'family flock' would be constituted and managed by a member of the family.

Characteristics of the flocks managed and feeding conditions

The size of the flock managed by the shepherds (Table 2) ranged from 100 sheep and 20 goats (J in El Ksabi) to 250 sheep and 15 goats (D in Boulemane). Larger flocks were divided into smaller units of 200–300 animals, which was the size considered to be manageable by one person. Such flock size corresponds to medium to large farms in the study area (Bechchari *et al.* 2014). All interviewees considered that keeping goats was not a profitable activity in itself. Goats were kept in the flocks for three reasons: their milk was used for family consumption, they were useful and efficient guides for the whole flock, and they stimulated the motion rhythm. The higher proportion of goats in a few flocks was explained by the interviewees as a way to make better use of the forage resources disregarded by sheep. The main sheep breed was the *Timahdite*, especially in Boulemane. This breed is considered to be adapted to cold winters, and ANOC recently launched a program to develop it. In El Ksabi, where climatic conditions are warmer, the *Beni Guil* breed was also found, at least in crossbred animals, as it is considered to be more resistant to heat.

Across all sites, rangelands were common land and each farmer had a defined area on which he could graze his flock. Bringing the flock to the rangeland was, above all, a way to keep the rights to the land and ensure access to water sources. Thus, especially when other feed sources were available, feeding the flock on the rangeland became a secondary objective compared with access to water. The region has suffered serious droughts in the last 20 years, and watering the flock was considered as a matter of survival. As a consequence, most interviewees spent much time discussing the problem of water and the daily grazing circuits were sometimes designed solely as a function of water availability.

Steppe rangeland was available at all three sites. Generally, the animals were grazed on the steppe, but alternative forage resources were used whenever available. Traditionally, transhumance was practiced in Boulemane to bring the flocks to oak woods during the winter, but this practice is decreasing. Still, 3/5 interviewees moved their flock to Ifrane or Guigou. A lower utilisation of rangeland seemed to be correlated with the local availability of other feed resources such as cheap concentrates or cereal stubbles that have become increasingly available with the expansion of agricultural land. Cereal stubbles were preferentially devoted to grazing cattle. When they were available for sheep, they were used as a complement to rangeland during the summer period. In Guigou, 2/3 of the farms used self-produced hay and in El Ksabi, 2/3 of the farms grazed in the orchards during autumn and winter.

Table 2. Characteristics of the flocks managed by the shepherds interviewed and of the feed resources available

Site	Boulemane				Guigou		El Ksabi				
Interviewee	A	B	C	D	E	F	G	H	I	J	K
Animals in the flock (S: sheep, G: goats)	250 S ^A (600 in winter)	250 S 10 G	200 S 50 G	250 S 15 G	200 S 30 G	200 S (= lambs)	150 S 30 G	200 S 50 G	200 S 50 G	100 S 20 G	200 S 30 G
Sheep breed	<i>Timahdite</i>	<i>Timahdite</i>	mainly <i>Timahdite</i>	<i>Timahdite</i>	mainly <i>Timahdite</i>	mainly <i>Timahdite</i>	mixed	mixed	<i>Timahdite</i> + <i>Beni Guil</i>	mixed	<i>Timahdite</i> + <i>Beni Guil</i>
Types of rangeland	steppe + woods ^B	steppe + woods ^B	steppe + woods ^B	steppe	steppe	steppe + woods	steppe + woods	steppe + woods	steppe	steppe	steppe
Period of rangeland utilisation (months)	9	12	12	7	12	12	9	12	12	4	10
Other forage resources		–	Cereal stubbles	Straw	Cereal stubbles	Stubbles	Hay and stubbles	Stubbles + hay, straw	Orchards	Straw + stubbles	Orchards + stubbles
Supplementation with concentrate (months)	7	≈12	4 ^C	4	–	12	7	3	5	8	7

^A Another flock of 350 heads (comprising weaned lambs) was herded by a seasonal salaried shepherd in spring and summer; in winter, the two flocks were grouped for transhumance to another rangeland in the hills.

^B Woods are not found near Boulemane but the shepherds or farmers practice transhumance to Ifrane.

^C With self-produced cereals.

The utilisation of a diversity of forage resources (rangelands, orchard undergrowth, woods) reduced reliance on both rangeland and concentrate.

Concentrate supplementation varied between flocks: from almost nothing (E) to systematic supplementation (B and F). As a matter of fact, supplementation heavily depended on local feed availability. The latter included the forage production on the rangeland, the availability of locally produced concentrate and the feed provided through government support schemes.

LEK of the shepherds interviewed

LEK about pastoral resources

Shepherds referred to plants with local names. Often, these local names were given on the basis of their major morphological features, which enabled us to associate a local plant with a botanical genus (e.g. 'tawarra' for *Poa* sp.) or a botanical species (e.g. 'harmal' for *Perganum harmala*). However, associating local plant names with Latin names from the botanical classification was not always possible. For example, the name 'shiha' was given to both *Artemisia herba-alba* and *Santolina rosmarinifolia*, which share the same round shape of their tufts and the same thin and branched shape of their leaves. Shepherds also used generic terms such as 'idjejiugen' (flowers) to indicate the plants bearing flowers, which were all considered as highly palatable for sheep.

Shepherds named rangelands after their topography and, in the case of even topography, after their main forage plant. In Boulemane and El Ksabi, rangelands were referred to as 'asfal al jabbal' (foothills), 'haafa' (hillside) or 'alemou' (steppe). Based on the main fodder plant, 'alemou' was further referred to as 'halfa' (*Stipa tenacissima*) or 'shiha'. In the rare cases where the two plants could be found together, the rangeland was referred to as 'halfa', probably because this plant is large and abundant, thus providing more forage. In Guigou, two types of rangelands were identified: 'saffs' (woods) and 'idmren' (foothills). 'Saffs' refers to the woods of 'karoush' (*Quercus rotundifolia*), which are located on the hills surrounding the town of Guigou.

Different associations of forage plants were reported by most shepherds as a function of rangeland type. For example, 'karoush' and 'azoukuni' (*Thymus ciliatus*) were commonly associated in the woods; 'halfa', 'shiha' and 'tawarra' (*Poa* sp.) in the steppes; 'casdir' (*Genista quadriflora*) and 'agarbaz' (*Noaea mucronata*) in the foothills. However, only the two oldest shepherds described patch scale associations of plants (B: 'halfa and tawarra are always together, and where there is shiha, there is no halfa'; C: 'lafoudfoulous can be mainly found at this time [in spring] and only here close to the water and there is a lot').

Most shepherds discussed the seasonal availability of plants or plant parts in relation to animal feeding choices. Shepherds paid attention to the flowering period as flowers were generally eagerly eaten by sheep. The stage at which the two most available plants were less eaten or rejected by sheep was also well known: for 'halfa' (*Stipa tenacissima*), the flowering period was to be avoided, whereas 'harmal' (*Perganum harmala*) became edible only when in a wilted stage.

LEK about animal feeding choices

The experiential knowledge of shepherds about sheep feeding selectivity and circumstantial choices was variable. It ranged from the simple mention of a few preferred plants or plant parts (4/11 shepherds) to a full description of sheep feeding preferences depending on the season, on plant associations within a grazing site or grazing circuit, or even on the time of day with reference to drinking time (4/11 shepherds). Sheep feeding choices were considered to be consistently different from those of goats in terms of spatial distribution (C: 'goats prefer to go on the slopes (. . .), while sheep stay down [the slope]'), in terms of plants selected (J: 'sheep and goats do not eat the same thing beyond halfa'), or in terms of plant parts eaten (B: 'for halfa, they [sheep] do not eat flowers and stems but goats eat them (. . .), and for azoukuni, they [sheep] adore flowers and leaves'). Unsurprisingly, the ranking of local plants in terms of sheep preferences depended on the type and composition of the rangelands used by each shepherd (C in Boulemane: 'tawarra is the species they [sheep] eat most. They look for it everywhere, even when it is close to the ground. Next, I would say they like shiha, azkoun and taghardei'; H in Guigou: '[on the foothills] animals like tinegmit, tawarra and azoukuni'; J in El Ksabi: 'they [sheep] like halfa, smar and talzazat').

Animal feeding selectivity was described as season-dependent, the consumption of a given plant being a function of its stage and availability, as well as the stage and availability of the other locally available plants (H: 'harmal is eaten only when it is dry and the seeds are quite appreciated at the beginning of winter. . . otherwise they [sheep] will never eat it'; D: 'they [sheep] eat halfa only in winter because there is no more tawarra and the ground is sometimes covered with snow [halfa is taller]'). The selectivity for certain plants was further described as being affected by circumstances such as the availability of other plants in the vicinity (H: 'agarbaz is eaten mainly by goats but sheep eat the tufts that are mixed with tawarra') or timing of drinking (E: 'when the animals finish drinking, they will eat only azoukuni and then they will go for tawarra and azir and the others').

LEK about vegetation dynamics

Changes in the rangeland vegetation were identified by almost all interviewees and described as multifactorial, including: (i) a decrease in the number of plant species (B: 'there are not so many plants on the rangeland. As you can see, there is mostly this kind [showing *Stipa tenacissima*]'); (ii) the replacement of preferred species by much less palatable ones (G: 'shiha [*Artemisia herba-alba* or *Santolina rosmarinifolia*] and tawarra [*Poa* sp.] have become rare (. . .) there are more thistles and harmal [*Perganum harmala*], which are useless'); and (iii) a decrease in the size of the palatable plants still available (D: '[the rangeland] is poorer than before (. . .) before the animals would not walk so much and there was more tawarra [*Poa* sp.], which has now become so small (. . .) azoukuni [*Thymus ciliatus*] was much larger and we used its wood to cook but now we can't find it anymore'; K: 'ten years ago, tawarra [*Poa* sp.] grew high like barley, while now we only have small tufts, very sparse'). The main cause of such changes was considered to be drought. Vegetation

dynamics were associated, in terms of consequences, with the decrease in the number of farms and the increase in the need for concentrate feed. Only one shepherd (C) did not complain too much about forage availability on the rangeland: based on his observations, he was confident that rangeland vegetation would grow back with the rains.

Perceived palatability and chemical composition of plants

Chemical composition of plants was estimated as a function of the NIRS spectra. In order to avoid imprecise predictions, the samples whose spectrum was too different (Mahalanobis distance >3) from the CIRAD database used to produce the equations were not included in the analysis. This led to the rejection of a few samples such as *Perganum harmala* (L.) seeds, one of the *Santolina rosmarinifolia* (L.) samples, and *Rosmarinus officinalis* (L.). In the end, out of the 40 samples analysed, 11 were discarded either for the low quality of NIRS predictions or because the interviews suggested that the plant or plant part was not eaten by the animals. Based on the samples kept for the analysis, we can assume that sheep feeding on local rangelands ingest a variety of plant parts with different chemical characteristics (Table 3): among the samples analysed, estimations range from 21–82 for *in vitro* digestibility, 5–18 for crude protein, 26–79 for NDF and 2–14 for ADL (in %DM). The concentration of polyphenols and, especially, condensed tannins was overall moderate although variable, with a maximum concentration in the leaves and twigs of *Quercus rotundifolia* (Lam.) (4% DM polyphenols and 1% DM condensed tannins).

The plants or plant parts analysed are grouped in Table 3 on the basis of the sheep feeding preferences described by the shepherds or the viewpoint of the shepherd who has to ensure a proper daily intake for the flock: (i) eagerly selected by sheep; (ii) eagerly selected after drinking; (iii) important to ensure the

daily feed intake; and (iv) ingested although not preferred. Such classification shows that palatability for sheep and perceived value by shepherds do not seem to be directly correlated with the chemical composition since chemical characteristics may vary within a group. However, the plants (or plant parts) eagerly selected by sheep all the time had a low ADL content ($\leq 9\%$ DM) and a low concentration in polyphenols and condensed tannins (0.7–1.0 and 0.1–0.3% DM, respectively) in common. The plants eagerly selected after drinking all corresponded to species reported to contain aromatic compounds (Aafi *et al.* 2009); unfortunately, we were not able to measure aromatic compounds in our study. The plants considered as important by shepherds to ensure the daily feed intake were characterised by a high NDF content (56–79% DM) and a relatively low digestibility (21–43% DM). Interestingly, the plants considered as not preferred were characterised by high protein content (14–16% DM), medium to high digestibility (52–82% DM) and moderate concentrations of polyphenols (1.2–1.9% DM).

Integration of LEK into grazing management

The design of the daily grazing circuits was generally highly dependent on the location of the water source and on the surface area and spatial distribution of the land on which the owner of the flock had grazing rights. Thus, only 1/11 shepherds interviewed (E) clearly described deliberate changes in his grazing circuits on a seasonal basis in order to take best advantage of the diversity of plants available on the rangeland. Another 5/11 shepherds cited changes in the daily grazing circuits in order to ensure sufficient sheep intake. The choices in grazing management, when explained, could be directly associated with the knowledge the shepherd had of flock behaviour (G: 'the days when I have them eat shiha, then I bring them on tinegmit. Otherwise, they would not eat enough (. . .) I think that tinegmit must have a better taste and encourages them to eat more [shiha] (. . .) [without tinegmit] they would eat less shiha but more azkoun and tawarra').

Table 3. Chemical composition of the main plants described during the interviews

CP, crude protein; NDF, neutral detergent fibre; ADF, acid detergent fibre; ADL, acid detergent lignin; TP, total phenolics; CT, condensed tannins; IVOMD, *in vitro* organic matter digestibility

Species	Local name	Plant parts sampled (selected by sheep)	Ash %DM	CP %DM	NDF %DM	ADF %DM	ADL %DM	TP %DM	CT %DM	IVOMD %OM
Plants (or plant parts) described as eagerly ingested by sheep										
<i>Poa</i> sp.	Tawarra	Leaves + stems	7.3	11.4	62.0	30.6	3.3	1.04	0.31	45.1
<i>Scorzonera pygmaea</i> (Sibth. and Sm.)	Tinegmit	Aerial parts	15.4	8.4	35.5	24.6	8.2	0.68	0.10	67.4
<i>Festuca</i> sp.	Azkoun	Leaves	5.6	7.8	62.7	30.9	2.7	0.82	0.22	52.0
Plants (or plant parts) described as eagerly ingested by sheep after drinking										
<i>Thymus ciliatus</i> (Desf.)	Azoukuni	Leaves, twigs and flowers	9.7	10.2	47.1	34.3	13.7	0.74	0.09	53.6
<i>Santolina rosmarinifolia</i> (L.)	Shiha	Leaves + spikes	7.2	9.8	54.6	37.5	8.9	1.08	0.13	47.2
<i>Artemisia herba alba</i> (Asso)		Leaves and twigs	10.8	17.8	42.7	27.2	2.4	0.97	0.33	62.8
Plants (or plant parts) described as important in order to ensure the total amount ingested										
<i>Quercus rotundifolia</i> (Lam.)	Karoush	Leaves + twigs	4.3	12.1	55.6	37.5	14.3	4.05	0.99	42.7
<i>Juncus maritimus</i> (Lam.)	Smar	Leaves + stems	8.3	7.2	65.8	35.6	4.3	1.09	0.34	39.9
<i>Stipa tenacissima</i> (L.)	Halfa	Leaves + stems	4.3	5.0	79.1	48.7	6.9	0.84	0.15	21.3
Plants (or plant parts) described as ingested by sheep although not preferred										
<i>Ranunculus</i> sp.	Lafoudfoulous	Leaves + flowers	9.6	13.6	38.1	21.4	6.5	1.89	0.80	72.7
<i>Perganum harmala</i> (L.)	Harmal	Leaves (partly wilted)	15.1	16.1	26.2	17.9	5.1	1.41	0.45	82.4
<i>Genista quadriflora</i> (Munby)	Casdir	Leaves + twigs	7.5	15.8	48.8	31.6	9.5	1.24	0.15	52.4

Shepherds' profiles in terms of individual LEK and management practices

A first group of 4/11 shepherds can be characterised by their extensive knowledge about animal feeding preferences (Table 4). Those shepherds who fed their flock on the rangeland 12 months a year and owned sheep (either in the flock or outside) were either (i) old shepherds who had grazed the same area or who had tended the same flock for several decades (B and C), or (ii) younger shepherds who were very motivated and willing to improve the system performance. For example, H had found an interesting ‘niche’ market and worked at improving animal production, while E projected himself into becoming a farmer-shepherd and tried to minimise supplementation. Those two younger shepherds probably benefited from a better education: H was used to working with ANOC, and E spoke good Arabic instead of the local Berber dialect.

A second group comprising 2/11 shepherds was characterised by its knowledge about plants, without apparent connection to animal feeding preferences. Nevertheless, these shepherds mentioned the need of associating, within the daily grazing circuit of the flock, vegetation patches bearing complementary plants, in order to make the animals eat their fill. The two shepherds in this group were salaried, had worked with different farmers and owned a few sheep outside the flock. Among them, G was not very talkative and K had been a labourer in town before coming back to his homeland and becoming a shepherd, which he had been doing for the past 15 years, changing farms over time. Both had shepherded in their childhood. We can thus assume that both shepherds had acquired a certain LEK in their childhood, probably by listening or watching elders. Apparently, their LEK was not fed by current observations about animal behaviour and would thus probably not evolve much over time.

A third group consisted of 5/11 shepherds who were moderately knowledgeable about plants and animal feeding preferences and did not talk about adapting their herding management. Tentative explanations for this include (i) having little personal interest in feeding the animals on rangeland resources, or (ii) being asked as a salaried shepherd to follow the instructions of the farmer to the letter. It can be noted that shepherds in this group were generally younger than in the

other groups. Among them, 'J' had been a shepherd since his childhood but now worked more in his olive orchards, leaving the flock to his children every afternoon. 'I' worked as a part-time shepherd, caring for sheep only during the grazing season. His apparently limited LEK makes him more a guardian of the flock than a shepherd feeding his flock on the rangeland.

Discussion

Consistency of our results with previous findings

Previous scientific knowledge about shepherding on rangeland

Previous studies with experienced sheep and goat herders carried out for over 30 years in France show that herding corresponds to the daily feeding management of the herd or flock, based on the motivational response of the entire group of animals to site-specific conditions. Consistent with the results of our study, experienced herders claim that an animal's tendency to select and eat a given plant or plant part mainly depends on the content and structure of grazing patches, which may vary during the year as a function of climatic conditions and utilisation. A major role of the herder, in this context, is to design the daily grazing circuit in order to boost the appetite and intake of the flock (Meuret and Provenza 2015a, 2015b). These grazing circuits include strategic points such as water sources, comfortable mid-day resting places, salt licks, etc. (Vall and Diallo 2009; Krätli and Schareika 2010; Ghorbani *et al.* 2013). In our study, due to the scarcity of water, the grazing circuits were strongly affected by the position of the watering point and by the time programmed by the community for watering each flock.

The adjustment of circuits throughout a season is generally designed to ensure both proper feeding and the multi-year renewal of pastoral resources. In our study, the objectives of renewing the resources or providing self-medication opportunities were not mentioned by shepherds, whereas they were clearly expressed in other studies (see, e.g. Ghorbani *et al.* (2013) in Northern Iran). The absence of a long-term resource management perspective in the Boulemane region may be attributed to the common status of the land and to relaxed traditional rules, which increase the competition for the existing

Table 4. Shepherds' profiles in terms of LEK and connection between LEK and grazing management

[illegible]

resource and reduce the concern about sustainable grazing (Bourbouze and Gibon 1999). Another possible explanation could be the acute problem of water scarcity, which has become the main concern of Moroccan farmers.

LEK about rangelands, plants and feeding preferences in pastoral contexts

Naming rangelands after their topography or dominant plant species, naming plants after their morphology and associating them with functions in the feeding system (e.g. they provide dry matter or have nutritional value or medicinal properties) was reported for shepherds in other pastoral contexts (Kenya and Namibia: Bollig and Schulte 1999; Burkina-Faso: Vall and Diallo 2009; Iran: Ghorbani *et al.* 2013). Plant identification and classification as a function of their palatability, which might change with season or physiological stage is also a classical feature of shepherds' LEK. In our study, the nutritional variables we measured to characterise the chemical composition of plants or plant parts did not seem to explain their status in terms of animal preference [as described by shepherds]. The shepherds interviewed also recognised that the preference for a certain plant is partly influenced by the sequence of plant mixing and access to water. Mixed diets are known to create complex synergies among a wide range of chemical compounds (Meuret and Provenza 2015a). Synergies among primary and secondary compounds within a mixed diet can enhance detoxification processes and thus increase the intake of apparently inedible plants (see the review of Provenza *et al.* 2003). In future studies, the comprehensive information available within the complete NIRS spectrum of a plant sample could be further analysed to take a wider range of chemical characteristics possibly associated with animal feeding selection of plants within a diversified meal into account (Silué *et al.* 2016). Volatile compounds such as essential oils, which were lost in our case during sample drying and were therefore not analysed, could also explain why certain plants are selected by sheep. For example, the essential oils of *Thymus ciliatus* are well known for their antibacterial properties and could be ingested for self-medication, a type of behaviour observed in other grazing environments (Villalba *et al.* 2014). The notions of circumstantial palatability and of complementary foods at pasture, which need further investigation in 'real' farming contexts, might explain why animals foraging in varied environments have been found to eat more than would have been expected based on the nutritional value of individual plants (Meuret and Provenza 2015a, 2015b).

In our study, the more knowledgeable Moroccan shepherds attributed distinct values to young and more nutritious plants or re-sprouted plant parts, and to mature and more fibrous plants or plant parts that nevertheless 'make up the bulk of the diet'. In Boulemane, a typical sheep diet would include *Santolina rosmarinifolia* (CP: 10%; NDF: 55%; and digestibility: 48% DM) and *Thymus ciliatus* (CP: 10%; NDF: 46%; and digestibility: 55% DM), providing small, more nutritious bites, and *Stipa tenacissima* (CP: 6%; NDF: 82%; and digestibility: 19%), providing massive and more fibrous bites. This confirms previous studies that have shown that most herders deliberately encourage their animals to mix young plants or plant parts [usually of small size] with mature and more fibrous ones

[usually larger] in their diet. Large plants enable the animals to take massive bites and to thus achieve a high instantaneous intake rate (Agreil *et al.* 2006). They usually display an unappealing chemical composition: high ADF content (such as *Stipa tenacissima*) or high polyphenol content (such as *Quercus rotundifolia*). Nevertheless, they can be strategically important when the daily grazing time is restricted by weather conditions or traveling time. The complementarity between small nutritious bites and massive fibre-rich bites enables animals to efficiently balance their diet and compensate for medium forage quality with higher intake (Meuret 1997).

Threats to pastoralism and associated LEK in the study area

Droughts, vegetation dynamics and their consequences

The shepherds interviewed frequently mentioned changes in the rangeland vegetation cover, mainly smaller plant size and lower vegetation density. Knowledge about vegetation dynamics at various space or time scales has been described for other pastoral communities (Mongolia: Fernandez-Gimenez 2000; Namibia: Bollig and Schulte 1999; Ethiopia: Abule *et al.* 2005). The vegetation changes are often qualified as degradation in the scientific literature, either in Morocco (Benbrahim *et al.* 2004) or in other contexts (Abule *et al.* 2005). The shepherds interviewed in our study cited frequent droughts as the main cause of vegetation changes, as did herders in other studies (Namibia: Bollig and Schulte 1999; Ethiopia: Solomon *et al.* 2007). A trend towards dryer and warmer climatic conditions was observed for Morocco in the second half of the 20th century, with several periods of below-average precipitation and a trend towards shorter return times of droughts (Born *et al.* 2008). Schilling *et al.* (2012) and Bechchari *et al.* (2014) also report an increase in the frequency of dry years in the last three decades, from two to three droughts every 5 years.

Overstocking or overgrazing is considered to be the main cause of vegetation changes in the equilibrium theory (Benbrahim *et al.* 2004; Davis 2005). This paradigm is much debated. For example, Perevolotsky and Seligman (1998) argue that, in the Mediterranean context, heavy grazing is a form of land use that can be qualified as both efficient and ecologically sound. In other studies based on ecological knowledge, overgrazing is ranked as the first factor of degradation (Abule *et al.* 2005) or the second factor after droughts (Solomon *et al.* 2007). The commonly held assumption that pastoral utilisation degrades rangelands may have prevented shepherds in our study from mentioning it, as a way to defend themselves from such accusations. In arid and semi-arid regions, climatic conditions (especially rainfall) are known to have a prominent impact on rangeland vegetation dynamics. Recent works suggest that in such contexts, both equilibrium (utilisation-driven) and non-equilibrium (environment-driven) dynamics co-exist (Vetter 2005). Thus, it is most probable that the steppe vegetation considered in our study was changing due to both droughts and heavy grazing. Nevertheless, the assumption of degradation due to overgrazing (Benbrahim *et al.* 2004) has led to conservation programs based on temporary or semi-permanent exclusions from grazing (Boutot 2011) in order

to preserve the multiple functions of these areas (rangeland, forest, biological habitat, landscapes) and to improve their long-term management (HCEFLCD 2007). In our study area, the Ifrane National Park was created in 2004 in order to secure the conservation of the cedar forest. Grazing is generally excluded in the park, with a system of financial compensation for the farmers belonging to a sylvo-pastoral association (Aziz *et al.* 2014). Grazing exclusions, combined with the increase in cultivated surface areas and the effects of droughts, decrease the local availability of pastoral resources and thus challenge pastoral farming systems by drastically reducing the options to adapt grazing management to seasonal and yearly hazards.

An uncertain future for ecological knowledge about grazing

The decreasing interest in shepherding, either due to the lack of pastoral resources or to better locally available job opportunities, is one of the factors explaining the lower LEK about rangeland plants and animal feeding behaviour observed for several interviewees. In this study, we identified 'new ways of being a shepherd': (i) part-time shepherd; (ii) shepherd changing farms every year in search of better wages; (iii) shepherd bringing his flock to the rangeland to ensure access to water but not to feed the animals; and (iv) farmer keeping a flock as a secondary activity. In such situations, shepherds might devote less time and attention to observing the vegetation, understanding plant-animal interactions or experimenting with new management options. As a consequence, those 'new ways of being a shepherd' might seem incompatible with the perpetuation and renewal of ecological knowledge about grazing management on rangelands.

Changes in the way an activity is practiced modify the associated LEK. Several studies report the consequences of the integration of smallholder farms into the market economy, which can lead either to erosion of ethno-botanic knowledge (Mexico: Benz *et al.* 2000; Canada: Turner and Turner 2008) or to the development of a new body of ecological knowledge (Reyes-García *et al.* 2007). From this perspective, we can hypothesise that LEK about grazing management in the Boulemane province is undergoing changes. Based on the results of focus groups held in 2012 with farmers and local stakeholders in order to design the regional plan (*Schéma directeur d'aménagement des parcours des plateaux et plaines Nord-Atlasiques*), Chattou (2014) argues that the changes in farming practices and traditional social structures have modified the figure of the shepherd and the transmission of local knowledge in North-Eastern Morocco. In terms of LEK, among the 11 shepherds interviewed in our study, we can distinguish between:

- (1) Situations where pastoral farming is still valued and rangelands are still seen as a major source of feed (e.g. interviewees B and E). In this case the farmer, or a close relative such as a brother or son, is involved in shepherding and in decision-making. During the daily grazing circuit, the shepherd devotes due attention to animal behaviour and tries out new management practices in order to improve the performance of his system. We can assume here that

LEK about grazing rangelands will be passed on to the new generations and enriched.

- (2) Situations where farms have evolved into more profitable agro-pastoral systems. In this case, if shepherding is entrusted to a salaried shepherd who changes every year and is not free to make his own management decisions, his motivation to observe and learn about the local conditions might be small and LEK might dwindle. Conversely, if the farmer is involved in shepherding and is eager to improve his system (e.g. interviewee H), the future of LEK might also depend on the ability of local extension services to value the farmer's LEK and to help him to enrich it with scientific knowledge.

Conclusion and perspectives

Our study produced an organised description of the LEK about pastoral resources, animal feeding choices, circumstantial palatability of grazed plants or plant parts and vegetation dynamics, in a pastoral area of Morocco. The type of LEK we identified in Moroccan shepherds is consistent with previous findings about herding communities elsewhere. The amount of LEK and its connection to grazing practices varied among the shepherds interviewed depending on the local agricultural context and on their 'way of being a shepherd': salaried worker strictly following the instructions of his employer, salaried family member with opportunities to have his own flock in the future, farmer-shepherd devoting time and attention to feeding his flock with natural forage resources. We identified situations where this LEK was connected to grazing practices, with synergies between the two, and other cases where the perpetuation and renewal of LEK is probably compromised. Our results confirm and refine previous findings about the consequences of recent changes in farming practices and social structures in North-Eastern Morocco. Despite the small number of farmers investigated, we can assume that our findings are representative of the regional diversity, as far as medium to large farms are concerned.

Based on the hypothesis that LEK about plants and animal behaviour is necessary to implement sustainable grazing management, our study raises the issue of finding ways to perpetuate 'pastoral' LEK. In this respect, decisive factors will be the contribution of shepherds to grazing management decisions and the ability of local extension services to assess LEK and then find means to foster its improvement and dissemination. In addition, our study focussed on LEK about rangeland vegetation and its utilisation. When shifting from pastoralism to agropastoralism in an agricultural context, stubbles and other resources from cultivated lands become of major interest. These resources modify the feeding systems and can be included in the daily grazing circuits. Thus, the development of LEK should consider the respective and combined features of rangelands and cultivated land for feeding the flocks. This LEK could be improved by current scientific knowledge about mixing and sequencing diets at pasture. The latter, in turn, could be refined by taking a wider range of plant chemical and morphological characteristics into account.

Conflicts of interest

The authors declare no conflicts of interest.

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